BUSINESSES MATHEMATICS 2019

NOTE:

i) This section consists of 10 part questions and all are to be answered. Each question carries one mark.

ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Time: 15 minutes Max. Marks: 10

SECTION 'A' (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

(i) If $\begin{vmatrix} 2 & x \\ 4 & 4 \end{vmatrix} = 0$, then the value of x is: *8 *2✓ *0 *10

(ii) When a series of payments of equal amount is made at equal intervals, each payment is known as:
* Perpetuity *investment *Annuity✓ *Interest

(iii) If $10^x = 1000$, then x is equal to: *-2 *2 *3✓ *10

(iv) The ratio of 2.5 feet to 30 inches is:
*1:1✓ *5:2 *1:3 *4:5

(v) 20% of 600 is: *20 *1200 *120✓ *102

(vi) The graph os second degree equation $y = 2x^2 + 3x - 2$ opens:
*Upwards✓ *downwards *to the right *to the left

(vii) $\sqrt[5]{x}$ is equal to *$x^{5/2}$ *$x^{1/5}$✓ *$x^{2/5}$ *x^5

(viii) If x:25::24:30, then the value of x is :
*2 *20✓ *120 *0.2

(ix) The x-intercept and y-intercept of the equation $\frac{x}{2} + \frac{y}{3} = 1$ are:
* 2 and 3✓ *3 and 2 *$\frac{1}{2}$ and $\frac{1}{3}$ *2 and $\frac{1}{3}$

(x) The binary equivalent of decimal number 7 is:
* 1110 *111✓ *1001 *101010

BUSINESSES MATHEMATICS 2019

TIME: 1 Hours 45 Minutes (40 Marks)

SECTION 'B' (SHORT ANSWER-QUESTION) (25 Marks)

Note: Answer any 5 questions from this section.

2. i) Compute compound interest on Rs. 37,000 for 4 years at the rate of 6% per annum.

ii. Perform the Following Operations to the binary numbers:

(a) 110010 − 11000 (b) 11001 × 1001 ÷ 111

iii) If $A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$, Then Find A^{-1} and show that $A \times A^{-1} = I$

iv) 9 men consume 120 kg of rice in 20 days. In How many days will 15 men consume 250 kg of rice?

v) Solve the following equation by Cramer's Rule:

$$2x + 2y = 18, \quad 3x - 3y = 21$$

vi) Find the equation of the straight line which passes through the points (10, 7) and (12, 9), also find the x-intercept and y-intercept of the straight line

vii) if $y = 1 + x^2 - 4x$, find the direction and Vertex of Parabola.

viii) Convert the decimal number 50 into its equivalent binary number and the binary number 110010 into its equivalent decimal number.

SECTION 'C' (DETAILED-ANSWER QUESTIONS)(15)

NOTE: Attempt any Two questions from this section.

3. If $A = \begin{bmatrix} 3 & 4 \\ -1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 5 \\ 2 & -3 \end{bmatrix}$, then find:

$(a) A + B$ $(b) A \times B$ $(c) A^t - B^t$

4. Solve the following equation: $\dfrac{3x+2}{2} + \dfrac{4x+6}{4} - \dfrac{2x-8}{8} = 16$

5. Find the sum of annuity and the present value of annuity if an amount of Rs. 20,000 is Invested at the end of each quarter for 4 years at 5% per annum compounded quarterly.

BUSINESSES MATHEMATICS 2021

(AS PER CONDENSED SYLLABUS)

TIME: 1Hour 30 minutes (Regular & Private) (75 Marks)

NOTE:

i) This section consists of 25 part questions and all are to be answered. Each question carries ONE marks.

ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

SECTION 'A' (Multiple Choice Questions)(25)

1. Choose the correct answer for each from the given options:

Choose the correct answer for each from the given options:

(i) The point (-2, 1) lies in this quadrant:
* 1^{st} * 2^{nd} ✓ * 3^{rd} * 4^{th}

(ii) Discriminant in the quadratic formula is:
* $\frac{b^2}{2a}$ * $b^2 - 4ac$ ✓ * b^2 * $\sqrt{b^2 - 4ac}$

(iii) If A is a singular matrix, then its inverse is:
* Identity Matrix * Null Matrix
* Scalar Matrix * Not Exist✓

(iv) The value of x : 5 :: 12 : 30 is:
* 6 * 0 * 3 * 2✓

(v) The equation x = 0 represents:
* x - axis * y - axis✓
* both x – axis and y - axis * Origin

(vi) A square matrix 'A' will be symmetric matrix if:
* $A^t = A^{-1}$ * $A = A^t$ ✓ * $A = -A^t$ * $A = I$

(vii) $(AB)^t$ =:
* AB * A'B' * BA * B'A'✓

(viii) The slope of the line 2x + 2y = 0 is:
* 1 * 0 * -1✓ * 2

(ix) The ratio of 8kg and 800g is:
* 8:16 * 1:20 * 80:15 * 1:10✓

(x) $(4)^{\frac{1}{2}} x (25)^{\frac{1}{2}}$ = :
* 2 * 10✓ * 100 * 6^1

(xi) The points where parabola cuts x-axis are called:
* vertex * roots * origin * average

(xii) 35% of 900 is:
* 315✓ * 326 * 320 * 316

(xiii) Multiplicative inverse of $-\frac{3}{4}$ is:
* $\frac{3}{4}$ * $\frac{4}{3}$ * $-\frac{4}{3}$✓ * $-\frac{3}{4}$

(xiv) The ratio $\frac{4}{9} : \frac{1}{3}$ in the simplest form is:
* 4 : 3✓ * 3 : 4 * 9 : 3 * 3 : 1

(xv) If AX = B then X is:
* BA^{-1} * AB * $B^{-1}A$ * $A^{-1}B$✓

(xvi) The distance between the points (1, -2) and (2, -1) passing through the straight line is:
* $\sqrt{2}$ ✓ * 2 * 1 * -1

(xvii) The parabola of the quadratic function $y = 2x^2 - 3x - 9$ opens:
* downward * upward✓ * towards right * towards left

(xviii) If cost price of an article is Rs. 150 and loss is Rs. 15 then the selling price is:
* Rs 165 * Rs. 135✓ * Rs. 150 * Rs. 15

(xix) If $\begin{bmatrix} -x & 4 \\ 3 & 9 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 3 & 9 \end{bmatrix}$, then x = :
* -2 * 4 * 2✓ * -4

(xx) This matrix $\begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}$:
* Singular✓ * Identity * Non-Singular * Row

(xxi) The slope of horizontal line is:
* 1 * 0✓ * ∞ * -1

(xxii) $(101)_2 - (100)_2$ =:
* $(11)_2$ * $(101)_2$ * $(01)_2$ ✓ * $(10)_2$

(xxiii) 8 x 20 ÷ 4 + 6 = :
* 10 * 40 * 46✓ * 160

(xxiv) The decimal number 2 is equivalent to the binary number:
* 10 ✓ * 100 * 1000 * 11

(xxv) The product of (x – y) and (x – y) is:
* $(x + y)^2$ * $(x^2 + 2xy - y^2)$
* $(x^2 - y^2)$ * $(x^2 - 2xy + y^2)$ ✓

SECTION 'B' (Short-Answer Questions) (15)

NOTE: Attempt any THREE part questions. All questions carries equal marks.

2. i) A shopkeeper sold a book at 20% more than its cost. What was the selling price if cost of the book was Rs. 320?

2. ii) Perform the following operations to the binary numbers.

$$(101011)_2 - (1101)_2 - (10111)_2$$

2. iii) Find the scope and distance between points $(5, 2)$ and $(-3, 8)$

iv) Solve by using Cramer's rule.

$$3x + 2y = 1$$
$$5x - 3y = 27$$

OR

2. iv) (OR) 50 labourers construct 15 houses in 18 days then how long 30 labours can construct 10 houses.

2. v) If $A = \begin{bmatrix} 4 & -1 \\ 3 & 3 \end{bmatrix}$ then verify $A + A^{-1} = 1$

2. v) (OR) If $\begin{bmatrix} 2 & 7 & 3 \\ -2 & 3 & 0 \\ -8 & x & -12 \end{bmatrix}$ is at Singular Matrix then find the value of 'x'.

SECTION "C" (Detailed Answer Questions)(10)

NOTE: Attempt any ONE part questions. All questions carries equal marks.

3. a) Divide Rs. 2410 among three brother Ashraf, Aslam and Akram in the ratio 2 : 3 : 5 respectively.

b) If $A = \begin{bmatrix} 2 & 3 \\ 4 & 4 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$, then Find $2A = 3B$

a) For quadratic function, $y = 3x^2 + 5x + 2$ find.

i) Which any parabola will open?

ii) Vertex

iii) Roots

BUSINESS MATHEMATICS 2018

TIME: 1 Hours 45 Minutes **(40 Marks)**

SECTION 'B' (SHORT ANSWER - QUESTION) (25 Marks)

NOTE: Answer any FIVE questions from this section.
Each question carries equal marks.

2 i) In what time will Rs. 200/- amount to Rs. 270/- at the rate of 5% per annum?

ii) If $X = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$, $Y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $Z = \begin{bmatrix} 3 & -10 \\ -6 & -1 \end{bmatrix}$, find the values of a and b where $aX + bY = Z$

iii) Solve for x: $\dfrac{2x+3}{2} + \dfrac{x}{4} = \dfrac{x}{6} - \dfrac{2x}{3}$

iv) Distribute a profit of Rs. 50,000/- among three partners Jawwad, Areeba and Humza, such that Jawwad and Areeba get the profit in the ratio 5:3 whereas Areeba and Humza get it in the ratio 3:2.

v) Solve the following equations using Cramer's Rule:

$$x + 2y = 6 \quad , \quad 2x + 7y = 3$$

vi) An item is sold for Rs. 1200/-. A profit 30% is obtained. Find the cost price of item and amount of the profit.

vii) Find the equation of the straight line which passes through two points (5,4) and (3,6)

viii) Perform the following binary operations:

a) 11110 + 111000 + 111100 b) 1111 x 110 – 111

SECTION 'C' (DETAILED-ANSWER QUESTIONS)(15)

NOTE: Attempt any Two questions from this section.
Each question carries equal marks.

3. a) 30 men build 15 rooms of equal size in 18 days. How long will 20 men take to complete 10 rooms of the same size?

3. b) Find the roots of equation $14 - 9x + x^2 = 0$

4. For the quadratic equation: $y = 2x(4x - 1) - 15$

Determine:

a) Which way does the parabola open?

b) The vertex of the parabola. c) The roots of the equation.

5. Find the sum and the present value of the annuity of Rs. 6000/- invested at the end of each year for 10 years at the rate of 8% compounded annually.

BUSINESS MATHEMATICS 2018

Time: 15 minutes **(Regular & Private)** **Max. Marks: 10**

SECTION 'A' (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

i) $(25)^{\frac{1}{2}} \times (8)^{\frac{1}{3}}$ is equal to: *10✓ *3 *200 *8

ii. The simple interest on Rs. 4000 for six years at 3.75% per annum is Rs: * 900✓ *956 *960 *1960

iii. If $10^x = 100$, then x is equal to: *10 *-10 *2✓ *100

iv. The decimal equivalent of the binary number 100 is:
* 4✓ *5 *100 *10

v. $A \times A^1$ is equal to: * Null matrix *A * I✓ * A'

vi. The ratio of 30 inches to 5 feet is:
* 2 : 1 *2 : 3 *1 : 2✓ *3 : 60

vii. The value of x in 4 : 3 :: x : 12 is: * 16✓ * ±16 *-6 *36

viii. The parabola $y = -3x^2 - 6x + 7$ opens:
* downwards✓ * upwards * towards left * towards right

ix. If $|A| = 0$, the matrix A is called:
* Non-singular * Singular✓ * Square *Zero

x. The slope of the straight line y = -4x + 8 is:
* 8 *4 *2 *-4✓